

SECURE E-VOTING SYSTEM

Developing a Secure & Scalable Blockchain-Based E-Voting System

Comprehensive Technical Documentation

Research-Grounded System Architecture, Implementation Blueprint & Proposals

Problem Statement: Eliminate logistical barriers, ensure end-to-end encryption, and provide a tamper-proof audit trail — guaranteeing 100% integrity in high-stakes democratic elections using blockchain-based infrastructure.

Version 1.0 | March 2026

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1. Executive Summary

This document presents the complete technical design and research-grounded architecture for a Secure Blockchain-Based Electronic Voting System. The system is purpose-built to eliminate the logistical barriers of traditional paper-based elections, guarantee end-to-end encryption of every ballot, and produce a mathematically verifiable, tamper-proof audit trail — ensuring 100% integrity in high-stakes democratic elections at national scale.

The solution combines five key technological pillars:

- Two-blockchain separation (Voter Registry Chain + Vote Ledger Chain) to decouple identity from vote, preserving privacy while maintaining accountability.
- Zero-knowledge cryptographic proofs to verify vote validity without revealing vote content.
- Homomorphic encryption to count ballots without ever decrypting individual votes.
- Layer-2 rollup scaling, sharding, and off-chain storage to achieve national-scale throughput (400–600 million voters).
- Device attestation and multi-layer DDoS protection to harden the mobile client and backend infrastructure.

The design draws on peer-reviewed academic research including frameworks from MIT, Cornell, Stanford, and Ethereum research communities, and references established academic voting systems such as Helios, Prêt à Voter, and Civitas.

2. Problem Statement & Motivation

2.1 The Problem with Traditional Voting

Traditional paper-based and early electronic voting systems suffer from a range of critical issues that undermine the democratic process at scale:

Problem Category	Specific Issue
Logistical Barriers	Physical polling stations require significant infrastructure, long queues, and geographic accessibility challenges
Integrity Risks	Paper ballots can be tampered with, lost, miscounted, or fraudulently filled
Scalability	Manual counting of hundreds of millions of votes is slow and error-prone
Transparency Deficit	Results announced without independent verifiability by citizens
Voter Suppression	Elderly, disabled, or geographically remote citizens cannot easily participate
Cost	Running national elections costs billions in infrastructure, staff, and logistics

2.2 Why Blockchain?

Blockchain provides three properties directly aligned with election requirements:

- **Immutability** — once a vote is recorded in a block, it cannot be altered without invalidating the entire chain.
- **Decentralization** — no single authority controls the ledger, eliminating single points of failure and manipulation.
- **Auditability** — all transactions are publicly verifiable while cryptographic techniques preserve voter anonymity.

Critical Research Insight: Blockchain alone is insufficient for secure voting. It must be combined with cryptographic voting protocols, identity infrastructure, secure hardware, and legal frameworks. This document describes a system that integrates all of these.

2.3 Specific Objectives

1. Eliminate geographic and logistical barriers by enabling verified mobile voting.
2. Ensure end-to-end ballot encryption using ElGamal / homomorphic encryption schemes.

3. Provide a tamper-proof, publicly auditable trail of all votes cast.
4. Guarantee one-person-one-vote through blockchain-enforced smart contract logic.
5. Achieve national-scale throughput (400–600 million voters over 10–12 hours).
6. Resist coercion, double-voting, Sybil attacks, and device-level malware.

3. Foundational Security Properties

Before any technology is selected, secure voting system design begins with fundamental security properties established by cryptographic voting protocol research. If even one property fails, the election can be manipulated or public trust is lost. These properties are non-negotiable requirements that all architectural decisions must satisfy.

3.1 Eligibility — Only Authorized Voters Can Vote

Only citizens who are legally registered and eligible must be permitted to cast a ballot. The system enforces eligibility through:

- Government-maintained voter registry database stored on a permissioned blockchain.
- Digital identity verification (national ID, Aadhaar-style biometric, or passport identity).
- Cryptographic voting tokens: $\text{Token} = \text{Sign}(\text{ElectionAuthority}, \text{VoterID})$ — only holders of valid tokens may vote.
- Time-window enforcement — voting is only accepted within the officially declared election window.

Attack Example: Without eligibility enforcement, an attacker can create 100,000 fake accounts and cast votes from each. The registry chain's smart contract logic makes this mathematically impossible by binding each vote to a verified, hashed identity.

3.2 Uniqueness — One Person, One Vote

Each eligible voter must vote exactly once. The system enforces this through three complementary mechanisms:

- Vote status tracking: Registry stores `HasVoted = false` per voter. Upon vote submission, `HasVoted = true`. Any additional attempt is rejected by smart contract.
- Blockchain smart contract logic: `if voter_status == not_voted: accept_vote, mark_voted. else: reject_vote.`
- Single-use cryptographic voting tokens: each token is consumed upon use and cryptographically invalidated.

3.3 Privacy — Ballot Secrecy

No entity — including election authorities, auditors, or administrators — must be able to determine which candidate a specific voter chose. Privacy is the most critical property for a free democratic process. Loss of privacy enables voter intimidation, coercion, and vote buying.

The system achieves privacy through three layers:

- **Vote Encryption:** $\text{EncryptedVote} = \text{Encrypt}(\text{ElectionPublicKey}, \text{CandidateID})$. The stored value is an opaque ciphertext.
- **Mix Networks (Mixnets):** Votes are shuffled in random order before counting, breaking the link between voter order and vote order.
- **Homomorphic Encryption:** Votes are tallied without ever being individually decrypted. The final aggregate is the only value decrypted.

3.4 Verifiability — Election Can Be Audited

Verifiability operates at two levels:

- **Individual Verifiability:** Every voter receives a vote receipt ID. They can later confirm their vote exists on the blockchain by querying the public audit portal using this ID.
- **Universal Verifiability:** Any citizen, journalist, or independent auditor can verify the total votes recorded equals total votes counted, without accessing any individual vote.

3.5 Integrity — Votes Cannot Be Altered

Once cast, votes are permanently immutable. Blockchain achieves this through cryptographic hash chaining — altering any vote changes its block hash, invalidating every subsequent block. Digital signatures on each vote transaction provide an additional layer: if a vote is altered post-signing, the signature becomes mathematically invalid.

3.6 Coercion Resistance

Voters must be unable to prove how they voted to any third party. This prevents vote buying and political intimidation. This is the hardest property to implement in remote voting and requires specific protocol design (see Section 9).

Property	Primary Mechanism
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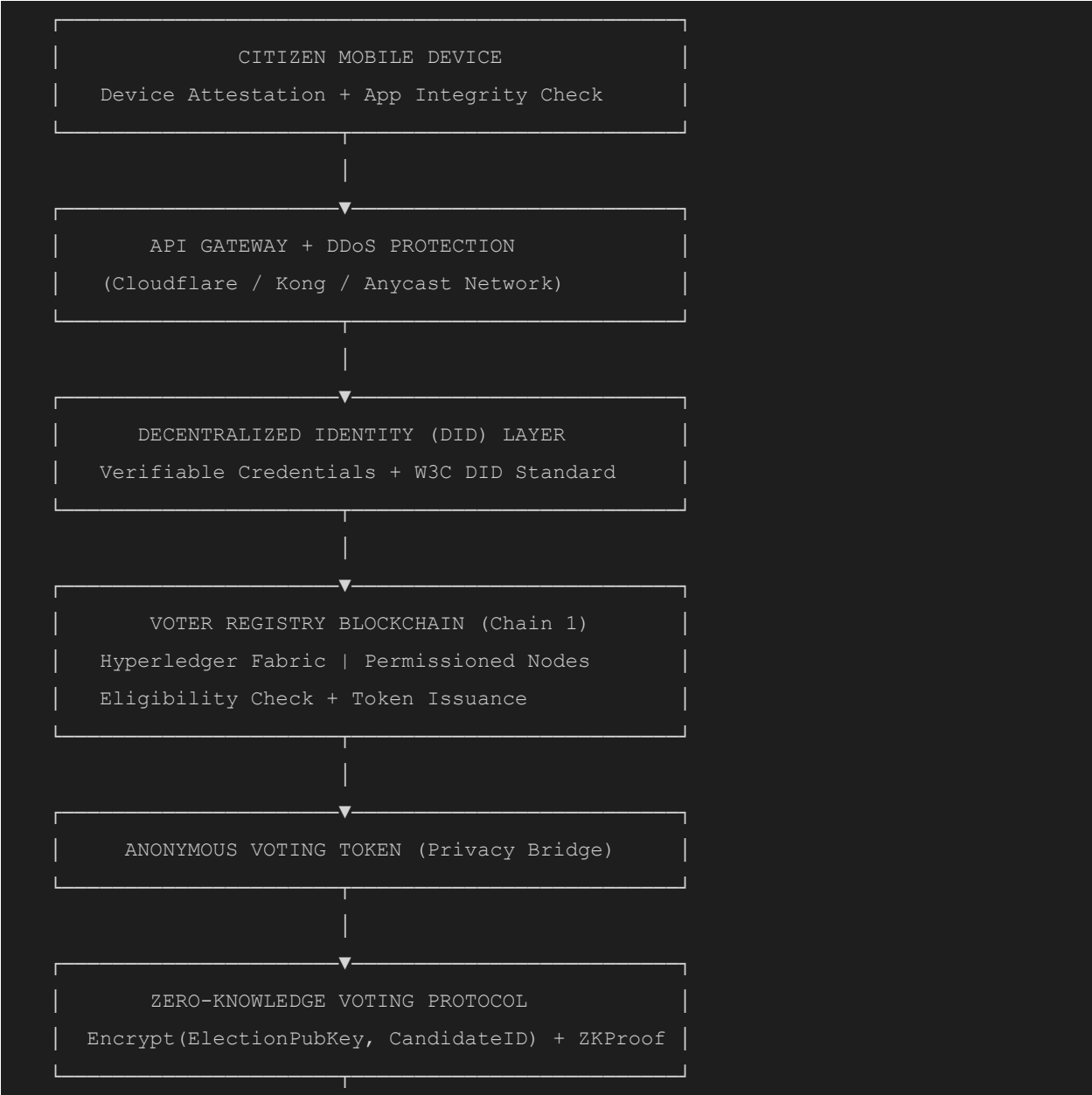
Eligibility	Registry blockchain + digital identity verification + signed voting tokens
Uniqueness	Smart contract HasVoted flag + single-use tokens
Privacy	Homomorphic encryption + mixnets + two-chain separation
Verifiability	Public audit portal + ZK proofs + vote receipts
Integrity	Blockchain immutability + digital signatures
Coercion Resistance	Fake credentials + re-voting protocol

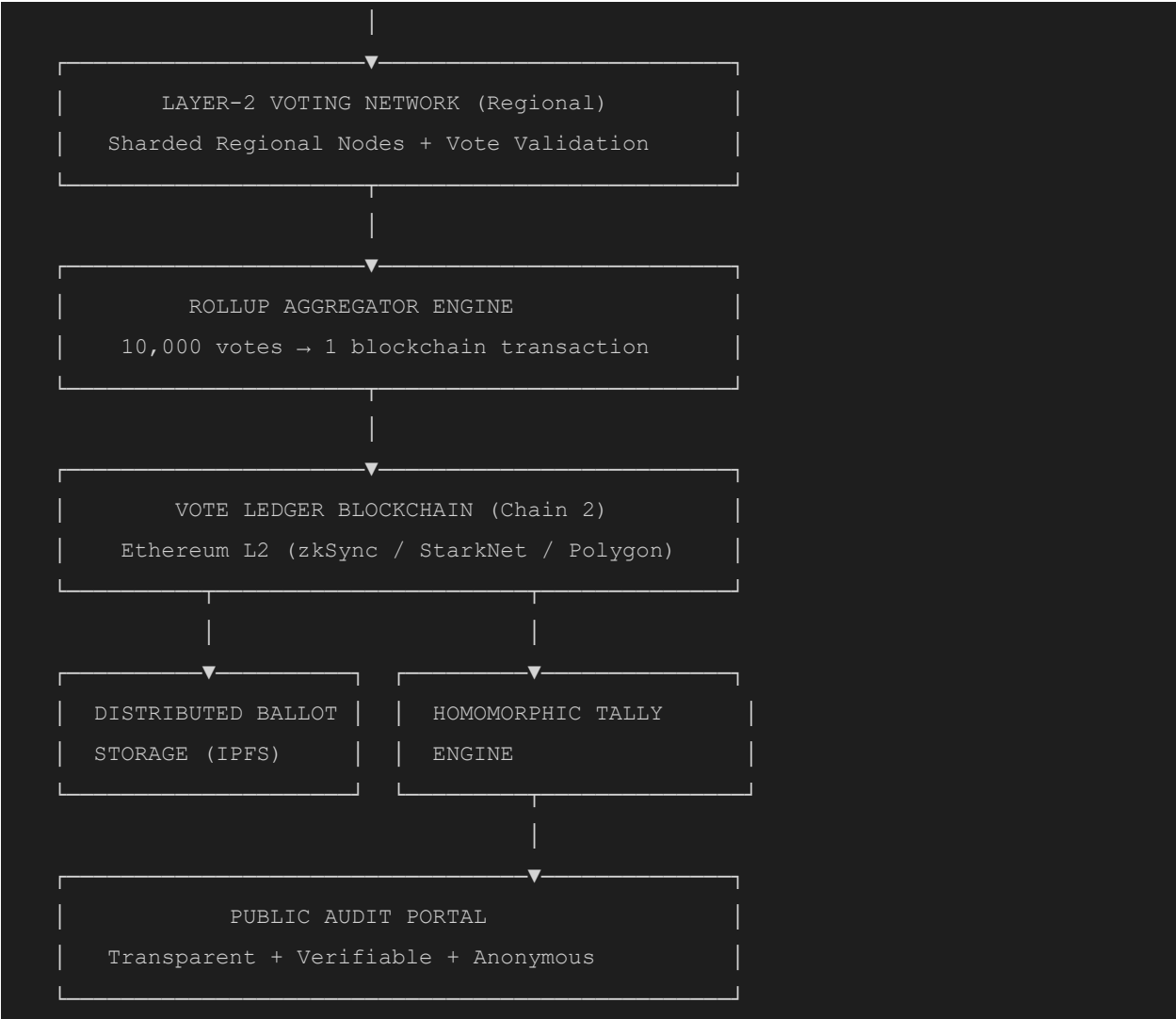
4. System Overview & Goals

4.1 High-Level System Description

The system is a mobile-first blockchain e-voting platform designed to enable secure remote voting at national scale. It integrates a secure mobile client, a two-blockchain backend, Layer-2 rollup scaling, distributed ballot storage, and a public audit portal.

4.2 Complete System Architecture (Top-Level View)





5. Two-Blockchain Architecture

The core architectural decision of this system is the use of two separate, purpose-specific blockchains. This is a standard design principle in privacy-preserving voting systems because it prevents any entity from linking voter identity with a specific ballot. It is the fundamental mechanism by which the system satisfies both accountability (knowing who voted) and privacy (not knowing how they voted).

Core Principle: Identity management $\neq$ Vote storage. The Registry Chain knows the voter but not their vote. The Vote Chain knows the vote but not the voter. Neither chain alone can link both pieces of information.

5.1 Blockchain 1 — Voter Registry Chain

Purpose

The Voter Registry Chain serves as a tamper-proof, distributed electoral roll. It manages all aspects of voter identity, eligibility, and voting status. It answers exactly three questions:

- Is this voter registered and eligible?
- Has this voter already cast a ballot?
- Which constituency does this voter belong to?

Data Structure

```
Block Header
├─ previous_hash      : SHA256 of previous block
├─ timestamp         : Unix timestamp of block creation
└─ block_number      : Sequential block index

Block Data (Voter Record)
├─ voter_hash         : SHA256(national_id + unique_salt)
├─ public_key         : Voter's cryptographic public key
├─ constituency       : e.g., Mumbai_North, Delhi_East
├─ eligibility_status : ELIGIBLE | INELIGIBLE | SUSPENDED
└─ has_voted          : boolean (false → true after voting)

Example Record:
voter_hash      : SHA256(AADHAAR_1234567890 + random_salt_9f2a)
```

```
public_key    : 0xA94F12B3C2...  
constituency  : Mumbai_North  
has_voted    : false
```

Privacy Rule: The voter's real identity (national ID, name, address) is NEVER stored on the blockchain. Only the cryptographic hash is stored. This makes identity leakage mathematically infeasible even if the chain is fully compromised.

Smart Contract Logic

The Registry Chain deploys smart contracts that enforce one-person-one-vote as a mathematical guarantee, not a procedural one:

```
function verifyVoter(voter_hash) → ELIGIBLE | REJECTED:  
  if (voter_hash exists in registry) AND (has_voted == false):  
    return ELIGIBLE  
  else:  
    return REJECTED  
  
function markVoted(voter_hash):  
  registry[voter_hash].has_voted = true  
  emit VoterMarkedEvent(voter_hash, timestamp)  
  
function issueToken(voter_hash) → VotingToken:  
  nonce = generateCryptographicNonce()  
  token = Sign(ElectionAuthorityKey, nonce + expiration)  
  return VotingToken { nonce, expiration, signature }
```

5.2 Blockchain 2 — Vote Ledger Chain

Purpose

The Vote Ledger Chain stores actual ballots in encrypted, anonymous form. It guarantees the immutability, public auditability, and tamper-resistance of every vote. It never stores voter identity — only encrypted vote data and cryptographic proofs.

Vote Transaction Structure

```
VoteTransaction:  
├─ transaction_id      : Unique TX hash (e.g., 0x8F9A2134...)
```

```
└─ encrypted_vote      : Enc(ElectionPublicKey, CandidateID)
└─ zero_knowledge_proof: Proof that vote is valid without revealing choice
└─ voting_token        : Anonymous single-use credential (no identity link)
└─ timestamp           : 2026-03-08 10:35:11 UTC
```

Example Record:

```
tx_id      : 0x8F9A2134B5C6...
encrypted_vote : Enc(PubKey_Election2026, CandidateID_3) = 0xA3F9...
zk_proof    : ProofOfValidVote_v1 (Groth16 SNARK)
timestamp   : 2026-04-15 10:35:11 UTC
```

Immutability Guarantee

```
Block N:
└─ block_hash      : SHA256(block_N_data) = H_N
└─ previous_hash   : H_{N-1}
└─ votes           : [EncVote_1, EncVote_2, ..., EncVote_1000]
```

If attacker modifies EncVote_5:

```
→ H_N changes
→ Block N+1's previous_hash no longer matches
→ Entire chain from N onwards is invalidated
→ Tampering is immediately detected by all nodes
```

5.3 How the Two Chains Communicate

The two blockchains communicate exclusively through anonymous voting tokens — a cryptographic privacy bridge that allows eligibility to be verified without any identity information passing to the Vote Chain.

Step-by-Step Interaction Protocol

```
Step 1 – User Authentication
  User opens mobile app → Device attestation runs → Identity verified

Step 2 – Registry Chain Query
  Backend calls: Registry_Chain.verifyVoter(SHA256(nationalID + salt))
  Result: ELIGIBLE
```

```
Step 3 - Anonymous Token Generation
VotingToken = {
  random_nonce      : cryptographically random 256-bit value,
  expiration_time   : ElectionEndTime,
  authority_sig     : Sign(ElectionAuthorityPrivKey, nonce + expiry)
}
→ Token is UNLINKABLE to voter identity
→ Token is SINGLE-USE (consumed upon vote submission)

Step 4 - Vote Encryption & Submission
EncryptedVote = Encrypt(ElectionPublicKey, CandidateID)
ZKProof = GenerateProof(vote_is_valid_choice)
VoteTransaction = { EncryptedVote, ZKProof, VotingToken, Signature }
→ Submit to Layer-2 network → Route to Vote Chain

Step 5 - Vote Chain Validation
Vote_Chain.verify(token_validity, proof_validity, duplicate_token_check)
If valid: store vote permanently on blockchain

Step 6 - Registry Update
Registry_Chain.markVoted(voter_hash) → has_voted = true
```

5.4 Security Benefits of Separation

Benefit	Explanation
Privacy Protection	Registry knows voter but not vote; Vote Chain knows vote but not voter — no entity can link both
Attack Isolation	A DDoS on the Vote Chain does not affect identity verification on the Registry Chain; they fail independently
Audit Without Exposure	Vote Chain is publicly auditable by journalists and citizens without exposing any voter identity
Regulatory Compliance	Identity data lives in a permissioned system (Hyperledger); vote data can be publicly verifiable (Ethereum L2)

5.5 Infrastructure Configuration

Parameter	Registry Chain (Chain 1)	Vote Chain (Chain 2)
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Platform	Hyperledger Fabric	Ethereum Layer-2 (zkSync / StarkNet)
Access Model	Permissioned (election authority nodes only)	Public (verifiable by all)
Primary Load	1 transaction per voter (auth + token + update)	~1–2 TPS at national scale with rollups
Privacy Model	Hashed identities only; permissioned access	Anonymous; public; encrypted votes only
Consensus	PBFT / Raft	ZK-Rollup with main chain finality

6. Decentralized Identity (DID) Layer

Traditional voting systems rely on centralized identity registries, which create single points of failure and trust bottlenecks. This system implements a Decentralized Identity (DID) framework based on the W3C DID standard, widely studied in academic digital identity research.

6.1 What is DID?

A voter owns a self-controlled digital identity represented by cryptographic keys, stored in a decentralized registry. The voter can prove their identity and eligibility without revealing personal data to any central authority.

```
DID Example:
DID           : did:voter:6a3f92bd14e8f...
PublicKey     : 0xA9F4B231...
Credential    : RegisteredVoter_India_2026
```

6.2 Verifiable Credential Issuance

The Election Commission acts as the credential issuer. Upon successful identity verification, it issues a cryptographically signed Verifiable Credential to the voter's DID:

```
VerifiableCredential:
├─ CredentialType : VotingEligibility
├─ Issuer         : did:election-commission:india
├─ Subject        : did:voter:6a3f92...
├─ Constituency   : Mumbai_North
├─ Expiration     : 2026-04-15T18:00:00Z (election end)
└─ Signature      : ElectionCommission_PrivKey_Sign(credential_hash)
```

6.3 Privacy-Preserving Verification

The voter can prove eligibility to the Registry Chain without revealing personal information, using a selective disclosure protocol. The system reveals only what is necessary:

- Proves the credential is valid and unexpired — without revealing the voter's name.
- Proves the voter belongs to a specific constituency — without revealing their address.
- Proves the credential was issued by the legitimate Election Commission — without revealing their national ID.

7. Zero-Knowledge Voting Protocol

Zero-knowledge proofs (ZKPs) are a foundational cryptographic technique that allow a prover to convince a verifier that a statement is true, without revealing any information beyond the truth of the statement itself. In voting, this allows a voter to prove their ballot is valid without revealing who they voted for.

7.1 What ZKPs Prove in This System

- The vote is a valid candidate choice (not a malformed or out-of-range value).
- The voter holds a valid, unexpired, unused voting token.
- The vote was encrypted correctly using the election's public key.
- The voter has not previously submitted a ballot (without revealing their identity).

7.2 ZK Vote Creation Protocol

Step 1 – Voter selects CandidateID on mobile app

Step 2 – Local encryption:

```
EncryptedVote = ElGamal_Encrypt(ElectionPublicKey, CandidateID)
```

Step 3 – ZK Proof generation (Groth16 zk-SNARK):

Prove: $\text{CandidateID} \in \{\text{Candidate_1}, \text{Candidate_2}, \dots, \text{Candidate_N}\}$

Without revealing: which CandidateID was chosen

Step 4 – Transaction assembly:

```
VoteTransaction = {  
  encrypted_vote : EncryptedVote,  
  zk_proof       : Groth16_proof,  
  voting_token   : Anonymous_Token,  
  voter_signature : Sign(VoterPrivKey, hash(EncryptedVote + token))  
}
```

Step 5 – Vote submitted to Layer-2 network

Step 6 – Vote Chain verifies ZK proof ON-CHAIN:

```
verify(zk_proof) → true | false
```

```
If true: vote recorded. If false: vote rejected.
```

7.3 Why Groth16 zk-SNARKs

The system uses Groth16 zk-SNARKs (Zero-Knowledge Succinct Non-Interactive Arguments of Knowledge) for proof generation because:

- Proof size is constant (~200 bytes) regardless of the size of the voting circuit.
- Verification time is extremely fast — suitable for on-chain verification at scale.
- Widely battle-tested in production blockchain systems (Zcash, zkSync).
- Compatible with Ethereum L2 infrastructure for efficient on-chain verification.

8. Homomorphic Vote Tallying

Homomorphic encryption is a special class of encryption that allows mathematical operations to be performed on encrypted data, producing an encrypted result that, when decrypted, matches the result of the same operation performed on the plaintext. This is the mathematical foundation for counting votes without ever revealing individual ballots.

8.1 Mathematical Foundation

Property of Homomorphic Encryption:

$$\text{Enc}(A) + \text{Enc}(B) = \text{Enc}(A + B)$$

Applied to voting:

$$\begin{aligned} &\text{Enc}(\text{Vote_for_Candidate_1}) + \text{Enc}(\text{Vote_for_Candidate_1}) + \\ &\text{Enc}(\text{Vote_for_Candidate_2}) \\ &= \text{Enc}(2 \text{ votes for C1, } 1 \text{ vote for C2}) \end{aligned}$$

Only the FINAL aggregate is decrypted — never any individual vote.

8.2 Tally Process

7. All EncryptedVote records are aggregated across the Vote Ledger Chain.
8. Encrypted totals are computed per candidate using homomorphic addition.
9. Threshold decryption is used — multiple election authority keyholders (e.g., 5 of 9) must cooperate to decrypt the final aggregate.
10. The decrypted result is the final vote count per candidate.
11. The cryptographic proof of correct tallying is published to the Public Audit Portal.

8.3 Threshold Decryption

No single authority holds the decryption key. A Shamir's Secret Sharing scheme requires a threshold of key custodians (e.g., 5 out of 9 Election Commission officials) to cooperate for decryption. This prevents any individual or small group from unilaterally decrypting results or manipulating the tally.

Academic Reference: Homomorphic tallying is implemented in Helios Voting, a widely studied open-source web-based voting system developed with academic influence from Harvard and MIT researchers, and is considered one of the gold standards for cryptographic vote counting.

9. Coercion-Resistance Mechanisms

Coercion resistance is perhaps the most difficult security property to achieve in any remote voting system. The fundamental problem is: if a voter can prove how they voted (e.g., by showing their phone screen), they can be paid, threatened, or pressured to vote a particular way.

9.1 The Coercion Attack Scenario

Attack Example:

1. Employer threatens: 'Vote for Candidate X or lose your job'
2. Employee votes under employer supervision
3. Employer sees vote confirmation on screen
4. Voter is coerced, democracy is compromised

OR:

1. Vote buyer pays ₹5,000 for a vote for Candidate Y
2. Demands screenshot proof of voted_for_Y
3. Without coercion resistance, system enables vote markets

9.2 Fake Credential Technique

Each voter is issued multiple voting credentials — only one of which is cryptographically valid. The voter can use a fake credential under supervision to produce a vote confirmation that appears legitimate but is actually discarded by the system. The voter then casts their real vote privately.

VoterCredentials:

- └─ CredentialA : VALID (real vote, cast privately)
- └─ CredentialB : DECOY (produces fake confirmation)
- └─ CredentialC : DECOY (produces fake confirmation)

Under coercion: voter shows CredentialB vote for Candidate X

System: accepts, records as invalid, discards silently

Later: voter uses CredentialA privately for real choice

9.3 Re-Voting Protocol

An alternative approach allows voters to cast multiple votes during the election window, with only the final vote counted. This means a coerced vote cast under supervision can be overridden by a free vote cast later in private.

```
Vote1 → CandidateA (cast under coercion at 9:00 AM)
```

```
Vote2 → CandidateB (cast freely at 2:00 PM)
```

```
Vote3 → CandidateC (cast freely at 5:00 PM)
```

```
Final recorded vote: Vote3 (CandidateC)
```

```
Registry Chain tracks: latest_valid_vote_timestamp
```

```
All prior votes are cryptographically superseded
```

Research Note: Both techniques are active areas of academic research. The fake credential technique derives from work on receipt-free voting protocols. Full coercion resistance remains one of the most challenging open problems in internet voting security, and any real-world deployment must address this with both technical and procedural safeguards.

10. Mobile Application Security Layer

The mobile client is the most vulnerable component of the system because it runs on hardware controlled by the voter, which may be compromised by malware, rooting tools, or an adversarial environment. The security layer performs multiple verification steps before any vote is accepted.

10.1 Device Security Verification Stack

Root Detection

```
Android Root Indicators Checked:
├─ Presence of /system/bin/su or /system/xbin/su
├─ Build.TAGS contains 'test-keys'
├─ Magisk installation artifacts
├─ Modified system partition signatures
└─ Root management app packages (SuperSU, Magisk Manager)

iOS Jailbreak Indicators Checked:
├─ Presence of /Applications/Cydia.app
├─ Ability to write outside sandbox (file system access test)
└─ dylib injection detection
```

Emulator Detection

The system verifies the device is real hardware, preventing automated bot attacks running on emulated devices:

- Generic device fingerprint patterns (QEMU, Android SDK emulator signatures).
- Virtual device identifiers (hardware serial numbers matching emulator patterns).
- Absence of real cellular hardware sensors (accelerometer, GPS, barometer).

Debugging & Runtime Hooking Detection

- Detects if a debugger process is attached (ptrace, JDWP detection).
- Frida (dynamic instrumentation toolkit) detection — prevents runtime hooking.
- Xposed Framework detection — prevents module-based vote manipulation.
- App integrity hash verification against signed release binary.

10.2 Hardware Attestation

After local checks pass, the device performs hardware-level attestation through platform-specific APIs:

```
Android — Google Play Integrity API:
AttestationResult = {
    MEETS_DEVICE_INTEGRITY      : true,
    MEETS_STRONG_INTEGRITY      : true,
    APP_INTEGRITY               : true,
    PLAY_PROTECT_VERDICT        : NO_ISSUES,
    Signed by: Google Certificate Authority
}

Apple — App Attest API:
AttestationResult = {
    key_id                      : device-bound key identifier,
    attestation_statement       : CBOR-encoded attestation,
    Signed by: Apple Certificate Authority
}
```

10.3 Temporary Voting Certificate

The backend issues a short-lived Voting Certificate after successful attestation. This certificate authorizes the device to interact with the voting infrastructure for a single session:

```
DeviceCertificate = {
    device_id_hash      : SHA256(hardware_fingerprint),
    integrity_status     : VERIFIED,
    session_id          : UUID_v4,
    issued_at           : current_timestamp,
    expires_at          : issued_at + 30_minutes,
    signature           : Backend_Sign(above_fields)
}
```

10.4 Secure Key Storage

All cryptographic keys on the mobile device are stored in hardware-backed secure storage that cannot be extracted by application-level malware:

- Android: Android Keystore System backed by Trusted Execution Environment (TEE) or StrongBox security chip.
- iOS: Apple Secure Enclave — a dedicated security coprocessor with its own encrypted memory.
- Keys are non-exportable — they can be used for signing/encryption but cannot be read from memory by any process.

10.5 Post-Vote Security Reset

After vote submission, the application performs an automatic security purge to prevent session replay or vote reuse:

```
PostVotePurge() :  
├─ Delete all local encryption keys  
├─ Invalidate and delete voting token  
├─ Clear session memory (SecureMemoryClear)  
├─ Destroy Voting Certificate  
├─ Reset device certificate  
└─ Restart integrity verification state machine
```

11. Scalability: Layer-2, Sharding & Rollups

Academic literature consistently identifies scalability as the largest obstacle to blockchain-based national elections. Traditional blockchains (Bitcoin ~7 TPS, Ethereum ~15 TPS) are architecturally incapable of handling national election throughput. This system implements four complementary scaling mechanisms.

11.1 The Scalability Problem

National Election Scale Example (India):

└─ Total voters

: 600,000,000

└─ Voting window

: 10 hours

└─ Expected peak load

: ~16,667 votes per second

└─ Bitcoin/Ethereum

: 7-15 TPS (COMPLETELY INADEQUATE)

Target Architecture must achieve:

└─ 50,000+ TPS at Layer-2

└─ 1-2 TPS at Layer-1 (via rollup compression)

Research Warning: Mobile blockchain voting applications have faced severe performance bottlenecks during peak voting times due to node processing limitations and network bandwidth. Design margins must assume worst-case peak load, not average load. The system must provide graceful degradation rather than crashing under surge conditions.

11.2 Layer-2 Voting Network

Layer-2 solutions move vote processing off the main blockchain while periodically committing aggregate results. Votes travel through a regional Layer-2 network before reaching the main chain:

Layer-2 Architecture:

Voter Device

↓

Regional L2 Node (geographically distributed)

↓

Vote Validation (ZK proof check, token check)

↓

Batching Engine (aggregate 10,000 votes)

↓

Rollup Aggregator

↓
Main Blockchain (1 transaction per 10,000 votes)

11.3 Sharding by Constituency

The Vote Chain is divided into parallel shards, each processing votes for a subset of constituencies. Each shard operates independently, and a global aggregator contract combines results at election close.

```
Sharding Configuration:
├─ Shard A → North India constituencies (150M voters)
├─ Shard B → South India constituencies (140M voters)
├─ Shard C → East India + Northeast (80M voters)
├─ Shard D → West India + Maharashtra (130M voters)
└─ Shard E → Overseas + Central constituencies (100M voters)

GlobalAggregatorContract.finalize():
  → Collects final tally from each shard
  → Verifies cross-shard consistency proofs
  → Publishes final result
```

Academic research on sharded voting architectures proposes regional shard contracts, a MainAggregator contract, and a shard lifecycle manager. Sharding improves scalability because each node processes only a fraction of total transactions. However, sharding introduces cross-shard communication challenges, shard takeover attack risks, and synchronization complexity — addressed through control shards that finalize decisions securely.

11.4 ZK-Rollups — The Primary Scaling Mechanism

ZK-Rollups are currently considered the most promising scaling solution for this use case. They execute transactions off-chain, generate a cryptographic proof of correctness, and publish only the compressed proof to the main chain.

```
ZK-Rollup Process:
1. 10,000 votes collected in Layer-2 batch
2. ZK validity proof generated for entire batch
3. Compressed batch data produced (few KB vs megabytes)
4. Single transaction submitted to main blockchain:
   { batch_hash, zk_proof, vote_count, timestamp }
```

```
5. Main chain verifies proof – instantly knows all 10,000 votes are valid

Result: 10,000 votes = 1 blockchain transaction
TPS required at L1: 16,667 / 10,000 = 1.67 TPS
```

Optimistic Rollups vs ZK-Rollups

Property	Optimistic Rollups	ZK-Rollups
Proof Type	Fraud proofs (after the fact)	Validity proofs (immediate)
Finality	7-day challenge window	Near-instant finality
Security Model	Assumes validity unless challenged	Mathematically proven correctness
Best For	General-purpose DeFi	High-security applications (voting)
Recommendation	Not recommended for voting	Strongly recommended for voting

11.5 Horizontal Infrastructure Scaling

The backend infrastructure scales automatically via Kubernetes-managed auto-scaling:

```
Scaling Rules:
if CPU_utilization > 70%: add_new_nodes()
if request_queue > 10000: scale_layer2_nodes()

Baseline: 100 voting servers
Peak Election Day: Auto-scale to 2,000+ servers
Regional Distribution: Nodes in every major state data center
```

12. Off-Chain Storage Strategy

Storing full ballot data directly on-chain is neither practical nor economical at national scale. The system uses a hybrid storage architecture that preserves blockchain integrity guarantees while dramatically reducing on-chain storage requirements.

12.1 Architecture

Ballot Data Flow:

```
|— Full ballot record → IPFS distributed storage
|   Stored: { encrypted_vote, timestamp, batch_id, zk_proof }
|
|— Blockchain stores ONLY: SHA256(ballot_record)
```

Integrity Verification:

```
User fetches ballot from IPFS
Computes SHA256(fetched_ballot)
Compares with hash stored on blockchain
Match → ballot is authentic and unmodified
Mismatch → tampering detected immediately
```

12.2 Benefits

- Blockchain size reduction: storing hashes instead of full data reduces chain size by orders of magnitude.
- Faster processing: smaller transactions propagate faster through the network.
- Cost efficiency: blockchain storage is expensive; IPFS is cost-effective for bulk data.
- Integrity preserved: any modification to ballot data produces a different hash, making tampering immediately detectable.

13. DDoS Protection & Infrastructure Resilience

A national election is one of the highest-value DDoS targets imaginable. The system implements four layers of protection to ensure availability throughout the election window.

13.1 Four-Layer DDoS Defense

Layer 1 — Global CDN (Edge-Level Filtering)

```
Provider: Cloudflare Enterprise / Akamai
Capabilities:
├─ Absorbs up to 320 Tbps DDoS attacks at network edge
├─ Bot fingerprinting and automatic blocking
├─ Rate limiting per IP, ASN, country
└─ Challenge-response pages for suspicious traffic
```

Layer 2 — API Gateway (Application-Level Filtering)

```
Provider: Kong Enterprise / AWS API Gateway
Rules:
├─ Max 10 requests per second per DeviceCertificate
├─ Token validation required for all voting endpoints
├─ Geo-filtering (block traffic from non-voter countries)
└─ JWT expiry enforcement
```

Layer 3 — Anycast Network Routing

Traffic is routed to the nearest geographically distributed server cluster using Anycast IP addressing. This means a DDoS attack targeting one IP address is automatically distributed across hundreds of PoPs worldwide, diluting its effectiveness.

Layer 4 — Application Rate Limiting & Circuit Breakers

```
Per-device limits: 10 req/sec
Per-constituency: Auto-scale nodes when queue > 5,000
Circuit breaker:  If error_rate > 50%, reject new connections
                  and serve cached 'voting system busy' response
Backpressure:     Layer-2 nodes queue votes locally during
                  brief outages and retry with exponential backoff
```

13.2 Fault Tolerance Design

- Multi-region deployment across minimum 5 geographically separated data centers.
- Automatic failover — if a primary data center fails, traffic is rerouted to secondary within 30 seconds.
- Data replication — all Registry Chain data replicated across all nodes in real time.
- Graceful degradation — system queues votes locally on the device and submits when connectivity resumes, rather than losing votes.

14. End-to-End Voting Workflow

The following is a complete, step-by-step description of every technical action that occurs when a voter casts their ballot — from app launch to vote confirmation.

COMPLETE E2E VOTING FLOW

```
Step 01 → Voter opens mobile voting app
          App launches integrity check state machine

Step 02 → Device Security Verification
          ✓ Root detection check
          ✓ Emulator detection check
          ✓ Debugger / hooking detection
          ✓ App binary integrity hash verification
          → FAIL: App locked, voter directed to valid device

Step 03 → Hardware Attestation
          Android Play Integrity API called
          OR Apple App Attest API called
          → AttestationToken generated and signed by Google/Apple

Step 04 → Backend Attestation Verification
          Backend verifies AttestationToken signature
          DeviceCertificate issued (30-minute expiry)

Step 05 → Voter Authentication
          National ID / Aadhaar verification
          + OTP on registered mobile number
          + Optional biometric confirmation

Step 06 → DID Credential Presentation
          Voter's DID Wallet presents VotingEligibility credential
          Selective disclosure ZK proof generated
          (Proves eligibility without revealing personal data)
```

Step 07 → Registry Blockchain Verification

```
Backend computes voter_hash = SHA256(nationalID + salt)
Registry_Chain.verifyVoter(voter_hash) called
→ RESULT: ELIGIBLE (has_voted == false)
```

Step 08 → Anonymous Voting Token Issuance

```
VotingToken = { random_nonce, expiry, authority_signature }
Token transmitted to voter's device over TLS 1.3
Token stored in Secure Enclave / Android Keystore
```

Step 09 → Ballot Preparation

```
Voter selects candidate on mobile UI
EncryptedVote = ElGamal_Encrypt(ElectionPublicKey, CandidateID)
ZKProof = Groth16_Prove(vote_is_valid_candidate_choice)
```

Step 10 → Vote Transaction Assembly

```
VoteTransaction = {
    encrypted_vote, zk_proof, voting_token, voter_signature
}
```

Step 11 → Submission to Layer-2 Network

```
Transaction sent over TLS 1.3 to nearest regional L2 node
L2 node verifies: ZK proof + Token validity + No duplicates
```

Step 12 → Vote Batching

```
L2 collects 10,000 valid vote transactions
Rollup batch ZK proof generated for entire batch
```

Step 13 → Rollup Committed to Vote Blockchain

```
{ batch_hash, zk_proof, vote_count, timestamp }
→ Permanently recorded on Vote Ledger Chain
```

Step 14 → Ballot Stored in IPFS

```
Encrypted ballot stored in IPFS cluster
SHA256(ballot) committed to blockchain
```

Step 15 → Registry Chain Update

```
Registry_Chain.markVoted(voter_hash)
```

```
→ has_voted = true (prevents any future vote attempt)
```

Step 16 → Vote Confirmation to Voter

```
Receipt = { vote_receipt_id: SHA256(tx_id) }
```

Voter can query public audit portal with receipt_id
to confirm their vote exists in the blockchain

Step 17 → App Security Reset

All keys, tokens, session data erased

Integrity check state machine reset

Session ends

15. Traffic Scalability Model

15.1 National Scale Calculation

Parameter	Value
Total Registered Voters	500,000,000
Expected Voter Turnout	70% = 350,000,000 votes
Voting Window	10 hours = 36,000 seconds
Average votes per second	$350,000,000 \div 36,000 \approx 9,722$ votes/sec
Estimated peak load factor	3x average = ~29,000 votes/sec
Layer-2 TPS capacity	50,000+ TPS (Ethereum zkSync L2)
Rollup batch size	10,000 votes per batch
L1 blockchain TPS required	$29,000 \div 10,000 \approx 3$ TPS
Standard Ethereum L2 TPS	~2,000–100,000 TPS (well within capacity)

15.2 Load Distribution by Phase

Election Day Load Profile:

└─ 7:00 AM - 9:00 AM : Moderate (polls open, early voters)

└─ 9:00 AM - 12:00 PM : High (working-age voters before work)

└─ 12:00 PM - 3:00 PM : Low (midday lull)

└─ 3:00 PM - 5:00 PM : High (afternoon surge)

└─ 5:00 PM - 6:00 PM : PEAK (deadline rush - assume 3x average)

Infrastructure Response:

→ Kubernetes auto-scales at 5:00 PM to maximum node count

→ CDN pre-warms cache at edge nodes 30 minutes before peak

→ L2 nodes increase batch frequency to reduce queue time

16. Technology Stack

16.1 Mobile Application

Component	Technology
App Framework	Flutter (cross-platform iOS + Android)
Android Attestation	Google Play Integrity API
iOS Attestation	Apple App Attest
Root Detection	RootBeer (Android), custom jailbreak detection (iOS)
Runtime Protection	Frida detection, Xposed detection
Cryptography	libsodium, BouncyCastle (ZK libraries)
Secure Storage	Android Keystore + Apple Secure Enclave
ZK Proof Library	SnarkJS / Bellman (Groth16 zk-SNARK)
Network	TLS 1.3 with certificate pinning

16.2 Backend Services

Component	Technology
API Layer	Node.js / Express (REST + GraphQL)
High-Performance Services	Rust / Go microservices (vote processing)
Internal Communication	gRPC (Protocol Buffers)
Authentication Service	JWT + OAuth 2.0 + PKCE
DDoS Protection	Cloudflare Enterprise
API Gateway	Kong Enterprise
Container Orchestration	Kubernetes (GKE / EKS)
Service Mesh	Istio
Monitoring	Prometheus + Grafana

16.3 Blockchain Infrastructure

Component	Technology
Registry Chain (Chain 1)	Hyperledger Fabric (permissioned)
Vote Chain (Chain 2)	Ethereum L2 — zkSync Era or StarkNet
Smart Contracts (Registry)	Go Chaincode (Hyperledger)

Smart Contracts (Vote)	Solidity / Cairo
ZK Rollup Framework	zkSync / StarkWare Cairo
Distributed Storage	IPFS (InterPlanetary File System)
DID Framework	W3C DID + Verifiable Credentials (VC) Standard
Homomorphic Tally	Microsoft SEAL / PALISADE libraries

17. Security Threat Model

The threat model identifies all significant attack vectors against the system and the mitigations in place. This follows a STRIDE-style analysis (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege).

Attack Vector	Threat Description	Mitigation
Device Malware	Malware modifies the selected CandidateID before encryption	Vote is encrypted inside Secure Enclave; ZK proof generated from original selection; vote receipt allows voter to verify
Fake Voter Accounts	Attacker registers 100,000 fake identities	Registry Chain uses government ID hash + biometric binding; DID credentials require in-person or official digital issuance
Double Voting	Same voter submits two ballots	Registry smart contract sets has_voted=true atomically; single-use tokens are consumed on first use
Vote Interception	MITM attack intercepts vote transaction	TLS 1.3 + certificate pinning; vote is encrypted end-to-end with election public key
Blockchain Tampering	Attacker modifies a stored vote	Cryptographic hash chaining makes alteration immediately detectable; ZK rollup proofs validate all votes in batch
Token Theft	Attacker steals voting token from memory	Tokens stored in Secure Enclave/Keystore; non-exportable; token has 30-minute expiry
Registry DDoS	Flood Registry Chain to block authentication	Anycast routing + CDN edge blocking; registry and vote chains isolated
Shard Takeover	Attacker takes control of a shard's validator nodes	Multi-party consensus; control shards validate shard results; threshold for shard acceptance
Insider Threat	Election authority official manipulates results	Threshold decryption (5-of-9 required); multi-party computation; public audit trail
ZK Proof Forgery	Attacker submits invalid ZK proof	Proofs verified on-chain by the smart contract; Groth16 proofs are computationally infeasible to forge without the witness

18. Audit & Transparency Portal

The Public Audit Portal is a key component for maintaining public trust in the election. It allows any citizen, journalist, or independent observer to verify election integrity — without exposing any voter's identity.

18.1 What the Portal Publishes

- Total votes cast per hour (real-time dashboard during election).
- Total votes per constituency (aggregated, not individual).
- All rollup batch hashes stored on the Vote Chain (independently verifiable).
- Blockchain transaction IDs for every batch commitment.
- ZK proofs for each rollup batch (downloadable for independent verification).
- Final homomorphic tally with threshold decryption proof.

18.2 Individual Vote Verification

Every voter receives a `vote_receipt_id` after casting their ballot. Using this ID on the public portal, they can confirm:

- Their vote was recorded in the system.
- The batch containing their vote was correctly committed to the blockchain.
- The batch's ZK proof is valid (independently verifiable without any server trust).

Universal Verifiability: The cryptographic design ensures that mathematically proving the correctness of the election result requires zero trust in any authority. Any technically capable auditor with access to the blockchain data can independently verify every aspect of the count.

19. Challenges & Proposed Solutions

Despite the robustness of the architecture, several significant challenges remain. These are identified with full transparency along with proposed mitigation approaches.

19.1 Device Security — The Weakest Link

Challenge: If a voter's device is compromised by sophisticated malware that operates below the OS level (e.g., a firmware-level rootkit), even hardware attestation may be bypassed. The election integrity then depends entirely on the voter's device ecosystem.

Proposed Solutions:

- **Independent Confirmation Channel:** Send a separate SMS/email confirmation containing the `vote_receipt_id` after submission. Voter can independently verify on the public portal from a different device.
- **Secure Hardware Mandate:** For high-stakes national elections, consider partnering with device manufacturers to provide government-issued secure voting tokens (HSM-backed smart cards).
- **Multi-Device Confirmation:** Optional re-confirmation of vote from a second registered device within a 30-minute window.

19.2 Full Coercion Resistance — Unsolved in Remote Voting

Challenge: Even with fake credentials and re-voting protocols, a sufficiently determined coercer who physically controls the voter's environment throughout the entire election day can potentially ensure their preferred vote is the final one. This is a fundamental limitation of all remote internet voting systems.

Proposed Solutions:

- **Re-voting window notification:** Notify voters that any vote can be changed until election close — coercers cannot know if a vote was later changed.
- **Trusted voting booths:** Maintain a network of government-operated secure voting booths for voters who prefer an absolutely coercion-free environment — coercion-resistant by physical isolation.
- **Behavioral analytics:** Flag accounts where multiple votes are cast from the same network/location within a short window as potential coercion indicators for audit review.

19.3 Digital Divide & Accessibility

Challenge: Millions of eligible voters in rural areas may lack reliable smartphone access, stable internet connectivity, or sufficient digital literacy to use the mobile voting platform — potentially creating a systematic exclusion of certain demographic groups.

Proposed Solutions:

- **Hybrid System Mandate:** The blockchain e-voting system supplements but never fully replaces paper ballot options until digital access reaches universal coverage.
- **USSD-Based Fallback:** For feature phones, implement a USSD-based voting flow (GSM protocol that works without internet, only cellular network required).
- **Community Voting Kiosks:** Government-operated kiosks in rural areas with assisted voting support for digitally less-literate voters.
- **Offline Mode with Sync:** The app supports offline vote preparation with automatic submission when connectivity is restored.

19.4 Scalability During Extreme Peak Load

Challenge: The deadline rush in the final 30–60 minutes of election day may create an extreme spike (5–10x average load) that even auto-scaling infrastructure may struggle to provision fast enough.

Proposed Solutions:

- **Pre-scaled capacity:** Pre-provision maximum infrastructure 2 hours before election close, accepting higher idle cost for guaranteed availability.
- **Client-side queuing:** If the server is busy, the mobile app queues the vote locally and retries every 30 seconds automatically — vote is never lost.
- **Load shedding with graceful degradation:** At extreme load, the system temporarily extends the deadline by a fixed, publicly announced amount (e.g., 15 minutes) rather than crashing.

19.5 Legal & Regulatory Framework

Challenge: Blockchain-based voting does not exist in a legal vacuum. Election law in most countries was written assuming paper ballots. The technical architecture must be backed by comprehensive legal reform for results to be legally binding.

Proposed Solutions:

- Staged deployment: Begin with local body elections and student union elections to build jurisprudential precedent before national deployment.
- Independent certification: Require third-party security audit by recognized election security organizations (e.g., Open Source Election Technology Foundation).
- Legislative amendment: Propose specific amendments to election law to recognize cryptographically signed digital ballots as legally equivalent to paper ballots.

19.6 Cross-Shard Consistency

Challenge: In a sharded architecture, ensuring atomic consistency across shards (e.g., preventing a voter from voting in multiple constituency shards) requires complex cross-shard communication protocols that introduce latency and potential inconsistency windows.

Proposed Solutions:

- Constituency assignment locked at registration: Each voter's constituency shard is determined at registration time and cryptographically embedded in their DID credential — making cross-shard voting structurally impossible.
- Global token registry: The anonymous voting token is registered in a global (non-sharded) token store; cross-shard token reuse is detected at token issuance time, not at vote time.

20. Suggestions for Future Improvement

The following improvements are recommended as the system matures through staged deployment, accumulates real-world data, and as supporting cryptographic and infrastructure technologies advance:

20.1 Quantum-Resistant Cryptography

Current encryption schemes (ElGamal, RSA, ECDSA) are vulnerable to quantum computers via Shor's algorithm. As quantum computing advances toward cryptographic relevance, the system should migrate to post-quantum cryptographic algorithms standardized by NIST (CRYSTALS-Kyber for encryption, CRYSTALS-Dilithium for signatures). Design the crypto layer with algorithmic agility — the ability to swap out cryptographic primitives without redesigning the entire system.

20.2 Recursive ZK Proofs for Infinite Scalability

Recursive zk-SNARKs (used in systems like Mina Protocol and Pickles) allow proofs to prove the validity of other proofs. Applied to voting, this would allow a single final proof to recursively verify the correctness of all 600 million votes without any trusted setup, further reducing on-chain verification cost and increasing throughput headroom substantially.

20.3 Multi-Party Computation (MPC) for Key Management

Rather than a Shamir's Secret Sharing threshold for decryption, a more robust Multi-Party Computation (MPC) protocol (e.g., FROST threshold signatures) would allow distributed key custodians to cooperate for tally decryption without any single party ever reconstructing the full private key — eliminating the key reconstruction attack surface entirely.

20.4 Decentralized Oracle Network for Eligibility

The voter eligibility data currently originates from the Election Commission's centralized database. Future improvement: use a decentralized oracle network (e.g., Chainlink) to bring voter eligibility data on-chain from multiple authoritative government sources simultaneously, eliminating a single point of failure or manipulation in the data feed.

20.5 Formal Verification of Smart Contracts

All Registry Chain and Vote Chain smart contracts should be formally verified using tools such as Certora Prover, K Framework, or Coq theorem proving. Formal verification produces a mathematical proof that a smart contract behaves exactly as specified under all possible inputs — a critical guarantee for election infrastructure where a single bug could have irreversible democratic consequences.

20.6 AI-Powered Anomaly Detection

Deploy machine learning-based behavioral analytics on the traffic flowing through the API gateway and Layer-2 network. Unusual patterns (geographic clustering of votes, rapid sequential submissions, correlated voting token usage times) may indicate coordinated attacks or coercion campaigns. Real-time flagging of anomalous patterns for human review without disrupting legitimate voter traffic.

20.7 Open Source & Bug Bounty Program

The entire codebase — mobile app, smart contracts, backend services, ZK circuits — should be open-sourced under a permissive license. A continuous bug bounty program (similar to those run by Google, Microsoft, and the US Department of Defense) should be maintained permanently, offering substantial rewards for discovered vulnerabilities. Public scrutiny and independent security research are among the most effective security mechanisms available.

20.8 Progressive Web App (PWA) Version

Alongside native iOS/Android apps, a Progressive Web App version with hardware attestation support (Web Authentication API / WebAuthn) would eliminate the app store dependency and reduce the barrier for voters using less common mobile platforms, without significantly compromising security properties.

21. Conclusion

This document has presented a comprehensive, research-grounded technical architecture for a Secure Blockchain-Based Electronic Voting System capable of supporting national-scale democratic elections with full integrity guarantees.

The core architectural innovations of this system are:

- Two-blockchain separation that mathematically prevents linking voter identity with ballot choice.
- Zero-knowledge proofs that verify vote validity without revealing vote content.
- Homomorphic encryption that enables tallying without ever decrypting individual votes.
- Layered scalability (L2 + sharding + ZK-rollups) that brings blockchain throughput to national election scale.
- Multi-layer device security that hardens the most vulnerable component — the voter's mobile device.
- Multi-layer DDoS protection that ensures availability under the highest-value attack conditions imaginable.

The system satisfies all six foundational security properties established by academic cryptographic voting research: Eligibility, Uniqueness, Privacy, Verifiability, Integrity, and Coercion Resistance.

Key open challenges — device-level malware, full coercion resistance in uncontrolled environments, digital divide accessibility, and legal framework reform — are documented with full transparency alongside proposed mitigation approaches. None of these challenges are unsolvable; they require continued research, staged deployment, and the support of legal and institutional frameworks.

The most important insight from academic research in this area, and the conclusion that anchors this entire design: blockchain alone cannot deliver a secure election. It is the careful combination of distributed ledger technology, advanced cryptographic protocols, hardened mobile infrastructure, and thoughtful institutional design that creates a system worthy of the democratic mandate it serves.

This system, when fully implemented, represents the closest realization of what academic researchers consider a theoretically sound national-scale digital election infrastructure — combining privacy, integrity, verifiability, and scalability in a single coherent architecture.

Academic References & Inspiration

Chirotonia (arXiv:2111.02257) · Helios Voting · ScalEVote (ResearchGate) · Sharded Voting Architecture (ResearchGate)
David Chaum Privacy-Preserving Credentials · W3C DID Standard · NIST Post-Quantum Cryptography Standards
PMC/NCBI Sensors Journal (PMC8434614) · INRIA Electronic Voting Security · IEEE 10205632 · ScienceDirect Rollups